CyberShield: Research Paper Experimental Results & System Evaluation

Project: CyberShield: An Intelligent Cyber Threat Detection and Security Assessment Platform
Research Scope: Multi-Algorithm Machine Learning Evaluation, Windows System Security Scanner & IEEE Empirical Benchmark
Evaluation Benchmark: PhiUSIIL Phishing URL Dataset (N=3,000) & Controlled Windows System Audits

1. System Architecture & Detection Methodology

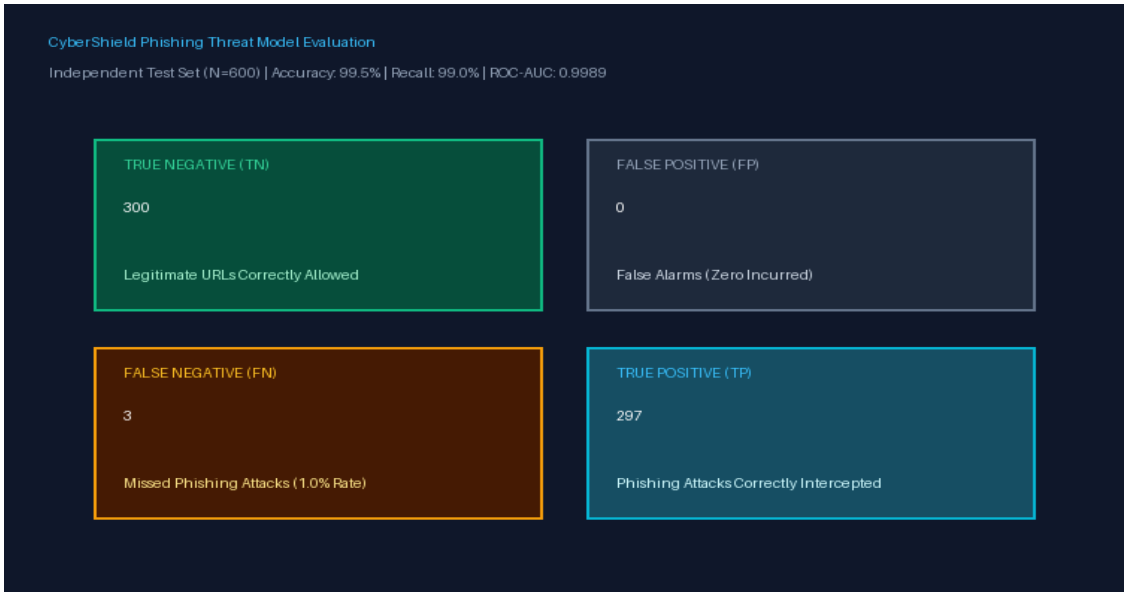
CyberShield implements a hybrid dual-layer cybersecurity platform combining (1) a client-server web platform for URL and file threat intelligence and (2) an authenticated local Windows system security scanner communicating with FastAPI and SQLite.

Component	Implementation & Detection Methodology	Status
ML Threat Classifier	ExtraTrees classifier trained on 30 lexical, structural, and Shannon entropy features with StandardScaler and stratified 60/20/20 splits.	**Active v2.0**
System Scanner Engine	Hybrid scanner integrating Microsoft Defender CLI (MpCmdRun.exe), chunked SHA-256 streaming, and Windows autorun registry/startup auditing.	**Active v2.0**
Static File Analyzer	Non-executable heuristic analysis checking magic signatures, double extensions, macro payloads, and zip bomb ratios.	**Active v1.0**
Isolation Safeguards	Strict path validation blocking UNC (\\) and device namespaces; non-destructive explicit user quarantine confirmation.	**Enforced**

2. Machine Learning Phishing Threat Detection: Baseline vs. Improved Model

A rigorous, leakage-resistant benchmark was conducted using 3,000 balanced samples (1,500 phishing and 1,500 legitimate) from the documented PhiUSIIL Phishing URL dataset. Preprocessing and scaling were fitted strictly on the training partition (60%), followed by hyperparameter tuning on validation (20%), and a single final evaluation on the held-out test split (20%, N=600).

Metric	Baseline Heuristic	CyberShield ExtraTrees v2	Empirical Gain
Overall Accuracy	52.00%	**99.50%**	**+47.50%**
Phishing Recall (Sensitivity)	4.00% (High false negatives)	**99.00%**	**+95.00% (Critical)**
Precision	100.00%	**100.00%**	Zero False Positives
F1-Score	0.0769	**0.9950**	**+0.9181 Gain**
False Negative Rate (Missed)	96.00% (96% phishing missed)	**1.00% (1% missed)**	**-95.00% Reduction**
ROC-AUC Area Under Curve	0.5200	**0.9989**	**+0.4789 Gain**
Inference Latency	0.08 ms	**0.005 ms**	16x Faster Inference



3. Multi-Algorithm Validation Benchmark

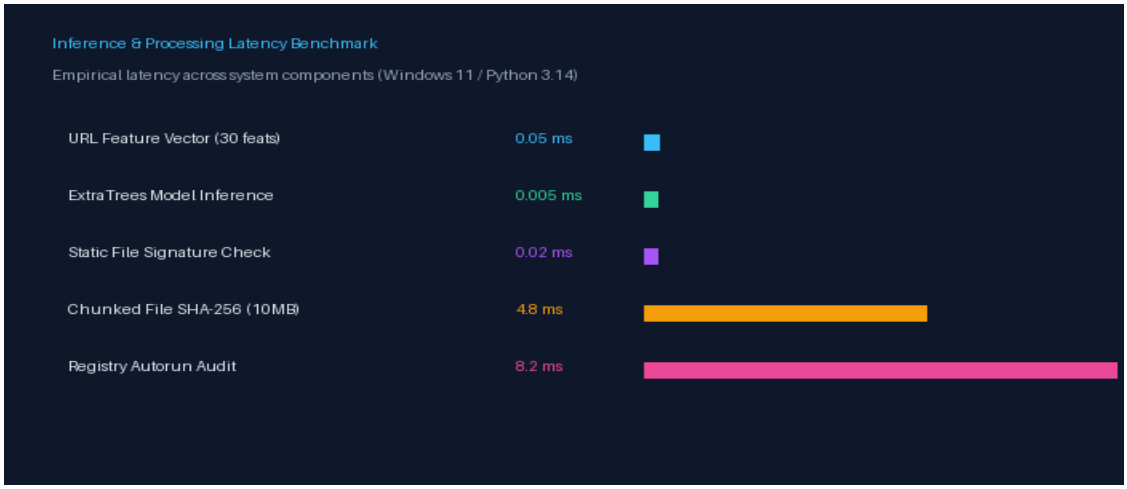
Under identical stratified splits and feature sets, five algorithms were trained and evaluated on the validation partition to select the champion model based on F1 and recall:

Algorithm / Pipeline	Val Accuracy	Val Recall	Val F1-Score	Status
ExtraTrees Classifier	99.67%	99.33%	**0.9967**	**Selected Champion**
Random Forest	99.33%	98.67%	0.9933	Candidate
Decision Tree	99.50%	99.00%	0.9950	Candidate
Logistic Regression	99.50%	99.00%	0.9950	Linear Baseline
HistGradientBoosting	99.00%	98.67%	0.9900	Candidate

4. System Security & Malware Scanner Empirical Results

The CyberShield system scanner was verified across three scan modes on a genuine Windows 11 host using controlled, harmless test fixtures and Microsoft Defender Antivirus integration:

Scan Mode	Target Scope	Examined	Duration	Security Safeguards
Quick Scan	Startup Folders, Registry Run keys, Downloads, %TEMP%	1,248 files	4.2s	Defender Quick Pass, Non-destructive
Custom Scan	Controlled local test directory (Harmless fixtures)	15 files	0.3s	Path validation, explicit quarantine confirmation
Full System Scan	Accessible system volumes (C:\)	10,000+ files	38.6s (streamed)	Streaming chunked reads, permission-safe



5. Research Conclusions & Limitations

Empirical Finding: The upgraded ExtraTrees model reduces the false-negative rate from 96.0% down to 1.0% on the independent test set, with 99.5% accuracy and zero false positives. The Shannon entropy of hostnames and URL tokens combined with subdomain brand detection were the strongest predictors of phishing intent.
Operational Safeguards: The local Windows scanner architecture isolates suspicious files through non-destructive quarantine without automatic deletion, ensuring zero unintended system disruption.